

Multi-source inputs

Flood labels
(DEP stormwater, 311,
USGS Ida HWM)

Static predictors
(terrain, flow-acc.
proxy, land cover,
hydro. distance)

Rainfall condition r
(constant synthetic;
not radar)

H3 assembly (R9)

Join layers to H3 cells

Provenance tags
(assembly · feature ·
label · rainfall)

Learning & validation

Gradient-boosting
classifier + continuous-
risk regressor

H3-block GroupKFold
spatial CV
(R7 parent blocks)

Logistic, ponding &
constant-class
baselines

Diagnostics & outputs

$PFI_h(c,r)$

Scale-loss Jaccard
ladder (R10 $\rightarrow$ R9 / R8)

Adaptive refinement
(PFI_h -guided $\rightarrow$ R11)

Sandy coastal-overlap
diagnostic

FEMA Sandy negative control
(never a training label)

